

An arithmetic barrier to sparse-profile integer blowups

Complete elementary proof; independently reviewed

30 September 2026

Status. Full independent mathematical review by the root agent passed; this is not formal verification.

A fair family with counts $c_1, \dots, c_n$ remains fair when every face of die i is replaced by b_i adjacent copies, for positive integers b_i . Its new count vector is $(b_i c_i)_i$; relabeling permutes these coordinates. A proposed equalization mechanism might first create profiles that are constant except on a few coordinates, then cancel the remaining noncommutative defects. The following elementary obstruction concerns that *first* step only.

Theorem 1. *Fix $n \geq 2$ and $1 \leq k \leq n$. There are fair families of arbitrarily large total size A such that every relabeled integer blowup whose count vector has at least k coordinates equal to one positive number M satisfies*

$$M \geq c_{n,F} A^k,$$

where F is the common count of any one fixed fair n -die family and $c_{n,F} > 0$ is independent of A . In particular, a method requiring a single profile of the form $M(1, \dots, 1) + (a_1, \dots, a_n)$ with at most q nonzero a_i cannot have a bound A^C uniform in n when $n - q > C$.

Proof. Start with a fair family having F faces on each die. For an integer $L \geq n$, blow up die i by $L + i$. The resulting counts are

$$c_i = F(L + i), \quad A = F \left(nL + \frac{n(n+1)}{2} \right) \leq 2nFL.$$

They are counts of an actual fair family, not merely an arithmetically admissible vector. If a further integer blowup, with any relabeling, has k counts equal to M , then for some k -element index set I ,

$$M \geq \text{lcm}_{i \in I} c_i = F \text{lcm}_{i \in I} (L + i).$$

For arbitrary positive integers $a_1, \dots, a_k$,

$$\text{lcm}(a_1, \dots, a_k) \geq \frac{\prod_i a_i}{\prod_{i < j} \gcd(a_i, a_j)}.$$

Indeed, at each prime the sum of all valuations except their maximum is at most the sum of the pairwise minimum valuations. Here $\gcd(L + i, L + j)$ divides $|i - j|$, and is at most n . Therefore

$$M \geq \frac{FL^k}{n^{k(k-1)/2}} \geq \frac{F}{n^{k(k-1)/2}} \left(\frac{A}{2nF} \right)^k.$$

This proves the explicit assertion. For the final claim fix any n, q with $k = n - q > C$ and let L tend to infinity. Any factor depending only on n is eventually dominated by A^{k-C} . $\square$

Exact scope. The obstruction applies to a single relabeling followed by coordinatewise integer blowups. It also applies to a scheme whose first nontrivial block must be such a near-constant profile. It does not exclude cancellation among many nonconstant profiles, rational continuous constructions, removing detectable redundant copies, or exploiting structural properties of a shortest family. In particular it is *not* evidence against a polynomial relation between the arbitrary-count and equal-count optima. The input examples deliberately contain redundant integer blowups.

Relation to positive splitting. For a fair family, its reduced signature is $\exp(\sum_i c_i X_i)$. A profile with only one exceptional coordinate would give the attractive direction $MH + aX_i$, where $H = \sum_i X_i$. The theorem shows why obtaining such directions by integer blowups alone can already cost almost the full generic least-common-multiple bound. A polynomial equalization argument using this idea would have to replace exact sparsity by a different mechanism or use information beyond the count vector and allowed blowups.